PROJECT AND TEAM INFORMATION

Project Title

TiffinGrid:A Hyper-Local, Graph-Optimized Tiffin Ordering and Delivery System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Students residing in PGs and accommodations around a university campus often struggle to find affordable, home-cooked, and timely meals. Existing food-delivery platforms primarily focus on commercial restaurants and cafés and generally do not provide features specifically designed for daily tiffin services, such as recurring meal schedules, daily/monthly plans, and convenient coordination with local home-food vendors.

There is also a privacy concern when students and delivery partners need to communicate, as directly exposing personal phone numbers may compromise user privacy.

The proposed project aims to develop a university-focused tiffin ordering and delivery system that connects students with local tiffin-service vendors. The system will provide vendor discovery, meal-plan management, time-slot scheduling, and optimized delivery routing within the university area.

The project will also incorporate a privacy-preserving communication mechanism using virtual numbers that allow businesses to automatically route incoming calls through a menu system. This will allow students and delivery partners to communicate without directly revealing their personal phone numbers.

The solution will demonstrate the practical application of Data Structures and Algorithms and Object-Oriented Programming in C++ to solve real-world problems involving searching, scheduling, routing, and secure communication.

State of the Art / Current solution

Currently, students generally depend on commercial food-delivery applications, individual local restaurants, or direct communication with nearby tiffin providers. Platforms such as conventional food-delivery applications are designed mainly around restaurants and one-time food orders rather than recurring home-cooked meal requirements.

Local tiffin vendors often manage orders and delivery schedules manually through phone calls or messaging applications. This can make it difficult to efficiently manage multiple customers, recurring orders, delivery priorities, and routes.

Therefore, there is an opportunity for a localized tiffin-management platform specifically designed for students around a university, combining vendor discovery, recurring meal scheduling, delivery optimization, and privacy-preserving communication in a single system.

Project Goals and Milestones

The primary goal is to design and implement a university-specific tiffin ordering, scheduling, and delivery management system using C++ Data Structures and Object-Oriented Programming.

Initial milestones:

- Develop a system for students to discover and select local tiffin vendors and available meal plans.
- Implement a Binary Search Tree (BST) for efficient vendor and menu searching.
- Create an OOP-based meal-plan hierarchy supporting daily tiffins and monthly subscriptions.
- Implement a Min-Heap/Priority Queue to prioritize deliveries according to time-slot urgency.
- Represent the university area as a graph using a dynamic adjacency list.
- Implement BFS algorithm to determine optimized delivery routes.
- Develop a privacy-preserving communication system using dynamically generated virtual numbers.
- Implement session management and exception handling for unauthorized or invalid communication requests.
- Test and benchmark the major data structures and algorithms.

The final milestones will involve system integration, testing, benchmarking, debugging, and demonstration of the complete ordering, scheduling, routing, and communication workflow.

Project Approach

The project will be implemented primarily using C/C++, with C++ being used for Object-Oriented Programming and Data Structure implementation.

The system is organized into multiple functional modules:

1. **Vendor and Menu Management:** A Binary Search Tree (BST) will be used to index vendors and menus, enabling efficient key-based searching and retrieval.

2. **Meal-Plan Management:** An OOP hierarchy will be created with a base `MealPlan` class and derived classes such as `DailyTiffin` and `MonthlyTiffin`, utilizing inheritance and dynamic polymorphism.

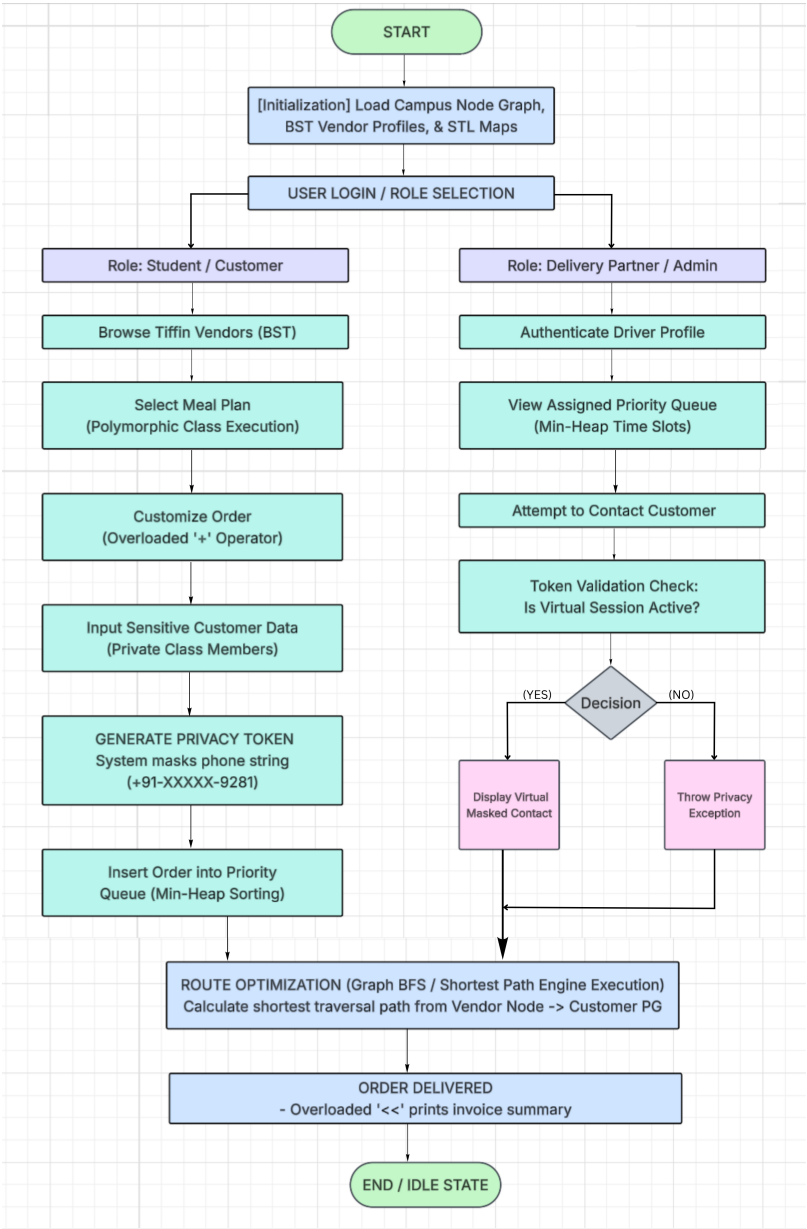
3. **Delivery Scheduling:** Incoming orders will be managed via a Min-Heap-based Priority Queue, prioritizing orders strictly by delivery time-slot urgency.

4. **Campus Route Optimization:** The campus and surrounding areas will be modeled as a graph using an adjacency list. Breadth First Search algorithm will determine optimal paths from vendors to customer residences.

5. **Security and Exception Handling:** Session verification checks will manage lookups, with C++ exception handling enforcing security against unauthorized or invalid sessions.

The complete system will be integrated into a modular architecture and benchmarked for algorithmic efficiency and correctness.

System Architecture (High Level Diagram)



Operational flow:

User Login / Role Selection → Vendor Discovery (BST) / Meal Plan (OOP) → Privacy Token Generation & Masking → Priority Queue Insertion (Min-Heap) → Campus Route Optimization (Graph Shortest Path) → Order Fulfillment.

The C++ core implements the data structures, graph routing engine, priority scheduling queue, and security session handlers.

Project Outcome / Deliverables

The project will deliver a working prototype of a localized university tiffin ordering and delivery management system.

The expected deliverables include:

- A vendor and menu search module using a Binary Search Tree.
- An OOP-based meal-plan system supporting daily and monthly tiffin subscriptions.
- A Min-Heap Priority Queue for delivery scheduling based on time-slot urgency.
- A campus graph representation using a dynamic adjacency list.
- Breadth First Search module for optimized delivery routes.
- Session management and exception handling for secure communication.
- Testing and benchmarking of the implemented data structures and algorithms.
- A final integrated prototype demonstrating the complete ordering, scheduling, routing, and communication workflow.

Assumptions

The system assumes that participating tiffin vendors operate within or around the university campus and provide predefined menus and meal plans. The university area is represented as a finite graph consisting of known locations and routes. Delivery partners have access to the system for receiving assigned deliveries. Meal orders and time slots are assumed to be entered correctly by users.

REFERENCES

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, *Introduction to Algorithms*, MIT Press.
2. Bjarne Stroustrup, *The C++ Programming Language*, Addison-Wesley.